

Problem Set 1

Directions

Submit two files on Moodle before the deadline:

- [1] your **solution set** as a *.pdf [2] the **R script** as a *.R type file

For both, the first line should include your name and the assignment number

If you worked with one or two other students write "group:" and then their names

[1] the **solution set** should include for each answer:

- the question number/part
- your answer in words
- any hand drawn diagrams
- the command in R that was used (if relevant)
- the output from Rstudio (if relevant)

[2] the **R script** should include:

- all the commands you used to generate the solutions
- individually written comments that explain your code to the grader

Alcohol consumption and Social Preferences

You have data from Zak et al. (2021) which studies the effect of alcohol consumption on cooperative behavior. The experiment has 128 participants who make decisions in the Public Goods Game after consuming drinks. Drinks either contain alcohol or a similar-tasting placebo. Participants were informed (before consenting to the experiment) that they would be randomly assigned to either the placebo (P) or the alcohol treatment (A). The assignment was only revealed after the experiment.

The Public Goods Game (PGG): each participant is endowed with \$10 and can contribute any amount between \$0 and \$10 (including \$0) to a common pool along with **three other** anonymous participants. The total contributions in the common pool are tripled and then split evenly among the four contributors. The final payout is the amount left from the endowment, plus any earnings from the split of the common pool.

Example:

Alice contributes \$4, Bob contributes \$2, Charlie contributes \$0, and Daniel contributes \$10. The total contribution to the pool is \$16, which is tripled and then split among the four participants. Each participant earns \$12 from the pool.

Alice takes home $(\$10 - \$4) + \$12 = \18 . Bob takes home $(\$10 - \$2) + \$12 = \20 .

Charlie takes home $\$10 + \$12 = \$22$. Daniel takes home $(\$10 - \$10) + \$12 = \12 .

Those who contribute benefit others in the group, but risk being taken advantage of by non-contributors ("free-riders"). An individual's earnings are highest when they free-ride by contributing zero as long as others in their group contribute nonzero amounts.

Research question:

What is the effect of alcohol consumption on free-riding in the public goods game?

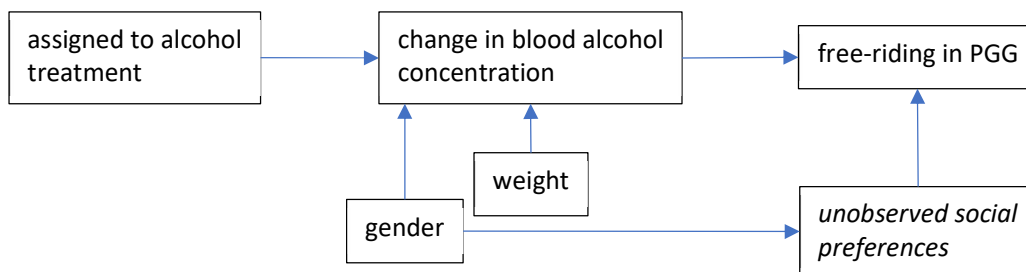
Dataset variables:

- ID – unique anonymous identifier
- age – age of the participant
- gender – gender of the participant (1 = male, 0 = female)
- oxytocin.chng – change in oxytocin level *
- stress.chng – change in stress (Adrenocorticotrophic hormone, ACTH) *
- blood.alcohol.chng – change in blood alcohol level (0.08 is legal driving limit) *
- pos.mood.change – change in positive mood (as measured through PANAS) *
- neg.mood.change – change in negative mood (as measured through PANAS) *
- PGG.contribution – amount (between 0 and 10) contributed to the common pool in the public goods game
- free.ride – indicator for contributing nothing to common pool in the public goods game (= 1 if contribution was 0, = 0 if contribution was greater than 0)
- alcohol.treatment – indicator if participant was randomly assigned to drink alcohol (=1) or the placebo (= 0)
- self.reported.intoxication – self-reported level of intoxication (0,..., 100)

* change is measured as the difference between the level after drinks minus the level at baseline before drinks.

Part I Practice Exam Questions

These questions should be completed without the use of AI as they will provide you with necessary practice for the exam.

Causal diagram

1. Given the causal diagram above, explain why we should use alcohol.treatment as our independent variable in the model instead of blood.alcohol.chng.

2. The snippet on the following page describes how the researchers administered the drinks to the participants. Each participant either received alcohol or a placebo non-alcoholic drink. Fill in the table listing everything someone in the alcohol treatment (A) experienced and the equivalent experience in the placebo (P). For experiences that were identical write “same” in the placebo column.

Alcohol administration

The protocol sought to induce a belief by participants' that they were consuming alcohol to mask the placebo even though the consent form stated that only one-half of participants would receive alcohol. This was done in several steps following previous protocols used in previous studies involving alcohol and social interactions [51,52]. This including have a bottle marked "vodka" on a drink cart where a research assistant mixed 1.5–3.0 ounces of 80-proof SKYY vodka (Gruppo Campari, Milan Italy) or an identical tasting placebo vodka (Arkay Alcohol Free Vodka, Jalisco, Mexico) with cranberry juice in clear highball glasses. Each drink consisted of one-part vodka or placebo, 3.5 parts cranberry juice. Males (females) received 62 (56) grams vodka per kilogram of body weight [53]. The expected BAC levels were 0.04–0.06%. Previous studies involving social alcohol consumption administered males (females) with 82 (74) grams vodka per kilogram body weight, resulting in BAC levels of 0.6–0.8% [54]. We reduced the amount administered to minimize the risk of any participant nearing the California legal driving limit of 0.08%. Participants were randomly assigned to the alcohol (A) or placebo (P) conditions. The rims of all glasses were dabbed with vodka in advance to induce a stinging sensation on the tongue to further influence P participants' beliefs they were consuming alcohol. These procedures have led participants in the placebo- condition to believe they had consumed alcohol in previous studies [55,56]. Alcohol metabolism varies with a person's weight and sex [57]. The drinks were mixed by the experimenters in front of participants and they were served one third of their drink every 10 minutes, over the course of 30 minutes. Participants had their BAC measured before being served subsequent portions of their drink. BAC levels were not shown to participants. After the conclusion of data collection, participants were not dismissed from the lab for safety reasons until their BAC reading was less than 0.03%.

Alcohol treatment (A)	Placebo (P)

3. Anything that varied between the control and treatment group is captured by "assigned to alcohol treatment" in the causal diagram. Referring to your table from the previous question, what varies between the placebo (P) and the alcohol treatment (A)?

4. Everything that is the same between the placebo (P) and the alcohol treatment (A) is an aspect of the experiment that the researchers wish to control for. The researchers wished to control for the perception of drinking alcohol. Discuss why the researchers needed to control for this and (referencing your table) how they achieved this.

5. Now consider only participants in the alcohol treatment (A).

- a. What would have happened to “change in BAC” if some of the participants drank the entire drink immediately and others drank it after 30 minutes?
- b. What would have happened to “change in BAC” if everyone were served 3 ounces of vodka, regardless of weight or gender?
- c. Why did the researchers serve the drink spread out over a 30 minute period but changed the amount of vodka served by weight and gender?

Part II Empirical Application

You are encouraged to train an AI agent to assist you with this portion of the assignment. Recall that you are responsible for any mistakes your agent makes. You must use critical thinking and judgement to decide if the answers it provides you are correct and if not direct it in correcting them.

6. For the following questions import the dataset into R. For each question write the formula for the model that you will use, justify its use, present your estimation results. State the conclusion you can draw based on a significance test or other hypothesis test.

- Verify that the alcohol treatment does not affect positive mood, negative mood, stress, or changes in oxytocin.
- Are participants in the alcohol treatment able to identify that they have had alcohol to drink more than those in the placebo? What does your model estimation tell us about the researchers’ ability to control for perception of drinking alcohol?
- What is the effect of alcohol consumption on free-riding? Use both an LPM and a probit model. Discuss the pros and cons of each of the models and whether the estimation results are robust to the different models.

Example of a correct answer for part 6:

- Verify that the alcohol treatment does affect blood alcohol level and that this effect does not vary with gender.

$$\text{blood.alcohol.chng}_i = \beta_0 + \beta_1 \text{alcohol.treatment}_i + \beta_2 \text{gender}_i + \beta_3 \text{alcohol.treatment}_i \cdot \text{gender}_i + u_i$$

We estimate the above model using OLS and find that effect of being in the alcohol treatment on change in blood alcohol level ($\hat{\beta}_1 = 0.058$, p-value < 2e-16), is statistically significantly different from 0 at the 5% level. We can conclude that being in the alcohol treatment increases the change in blood alcohol level by 0.058 points, which is within the upper range that the researchers aimed for (between 0.04 and 0.06 according to the snippet above). We see that this effect does not vary with gender since $\hat{\beta}_3 = -0.0059$ with a p-value of 0.403, which is not statistically significantly different from 0.

```
> summary(lm(blood.alcohol.chng ~ alcohol.treatment + gender + gender*alcohol.treatment, mydata) )
```

```
Call:
lm(formula = blood.alcohol.chng ~ alcohol.treatment + gender +
    gender * alcohol.treatment, data = mydata)

Residuals:
    Min       1Q   Median       3Q      Max
-0.032630 -0.006904  0.000000  0.000000  0.106273

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -1.496e-17  3.091e-03   0.000    1.000
alcohol.treatment  5.763e-02  4.832e-03  11.926 <2e-16 ***
gender         3.163e-18  4.945e-03   0.000    1.000
alcohol.treatment:gender -5.902e-03  7.039e-03  -0.839    0.403
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0193 on 120 degrees of freedom
(4 observations deleted due to missingness)
Multiple R-squared:  0.6732,    Adjusted R-squared:  0.665
F-statistic: 82.41 on 3 and 120 DF,  p-value: < 2.2e-16
```